AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

Design Phase – Solution 3

Project Focus: Automated Incident Classification | Technology: ServiceNow Flow Designer

1. Design Purpose

This design defines how an automated ticket-classification mechanism can be implemented for a school or college IT helpdesk. The solution uses ServiceNow Flow Designer to analyze incident information and assign a suitable Category and Subcategory. The design is intended for common requests such as Wi-Fi problems, projector failures, password issues, and slow computers.

2. Existing Process vs. Proposed Process

Existing Manual Process	Proposed Automated Process
User creates an incident and support staff manually identify the issue type.	User creates an incident and the flow automatically evaluates the request.
Category/Subcategory selection depends on manual review.	Predefined rules select the classification consistently.
Repeated tickets require repeated classification work.	Common patterns are processed automatically.
Unclear classification may require additional review.	Unmatched requests follow a defined fallback path.

3. Design Scope

- Incident creation and classification of common IT support requests.
- Evaluation of Short Description and Description fields.
- Rule-based selection of Category and Subcategory.
- Automatic update of the Incident record.
- Fallback handling for unmatched requests.
- Testing and validation of the complete flow.

4. Key Design Assumptions

- The required Category and Subcategory choices are available in the Incident process.
- Incident records contain meaningful text in Short Description and/or Description.
- The Flow Designer user has sufficient permissions to create, test, and activate the flow.
- Classification rules are agreed before final testing.

5. Solution Architecture

Input	Processing	Output
Incident record	Flow trigger and data retrieval	Incident available for classification
Short Description + Description	Keyword/condition evaluation	Matching issue type
Classification rules	Decision branches	Category + Subcategory
Updated Incident	Validation / completion	Classified ticket

6. Design Principle

The flow should perform one clear responsibility: identify the incident type and apply the correct classification. Assignment, notification, escalation, and other actions can be added later without changing the core classification concept.

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7. Detailed Flow Design

The automation is designed as a sequence of controlled stages. Each stage has a specific purpose, making the flow easier to understand and troubleshoot.

Step	Flow Stage	Design Action
01	Trigger	Start when a new Incident is created.
02	Read Record	Use the triggering Incident as the input record.
03	Evaluate Text	Inspect Short Description and Description for relevant terms.
04	Decision	Select the matching classification branch.
05	Set Values	Prepare Category and Subcategory values.
06	Update Record	Write classification values to the Incident.
07	Complete	End successfully and retain the updated ticket.

8. Classification Decision Model

The decision model uses a controlled set of conditions. Conditions should be ordered from specific issue patterns to general patterns. A final Else branch ensures that the flow has a safe outcome when no known keyword is detected.

Branch	Condition Pattern	Result
A	wifi / wireless / network / internet	Network → Wi-Fi / Connectivity
B	projector / display / HDMI / projection	Hardware → Projector
C	password / login / reset / credentials	Access → Password / Account
D	slow / lag / freeze / performance	Hardware → Performance
E	No matching condition	General → Other

9. Data Mapping Design

Source Data	Used For	Target Field
Short Description	Primary issue identification	Category / Subcategory
Description	Additional context for matching	Category / Subcategory
Matched rule	Classification decision	Incident Category
Matched rule	Classification detail	Incident Subcategory

10. Maintainability Design

- Keep each issue type as a clearly identifiable condition branch.
- Use meaningful labels for flow actions and decision steps.
- Document newly added keywords before changing the active flow.
- Test existing issue types after adding new rules to avoid unexpected matches.

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11. Validation Strategy

Validation is performed at three levels: rule validation, flow execution validation, and final incident-record validation. This confirms that the right condition is selected and that the resulting classification is actually stored in the Incident record.

Validation Level	Check	Success Criteria
Rule	Keyword matches the intended branch.	Correct branch selected.
Flow	All actions execute in sequence.	Execution completes without error.
Record	Category and Subcategory are updated.	Expected values appear on the Incident.
Fallback	Unknown request is processed.	General/Other is applied.

12. Test Scenarios

ID	Test Input	Expected Output
TS-01	“Classroom Wi-Fi is not connecting.”	Network → Wi-Fi / Connectivity
TS-02	“Projector has no display.”	Hardware → Projector
TS-03	“Please reset my login password.”	Access → Password / Account
TS-04	“Computer is slow and freezes often.”	Hardware → Performance
TS-05	“Need help with another IT issue.”	General → Other

13. Risk and Control Design

- Incorrect match: Review rule priority and make conditions more specific.
- No match: Use a fallback classification and allow manual review.
- Multiple keywords: Define a fixed branch priority before deployment.
- Flow failure: Review Flow Designer execution details and validate field values.
- Rule changes: Test existing scenarios again after modifying the flow.

14. Expected Deliverables

- Configured Incident classification flow in ServiceNow Flow Designer.
- Defined keyword-to-classification mapping.
- Test cases and expected results.
- Validated Incident records showing automatic classification.
- Project demonstration showing the complete automation lifecycle.

15. Future Extension

The same design can be extended after the core classification is stable. Possible extensions include automatic Assignment Group selection, priority suggestions, notifications to support teams, classification reporting, additional issue categories, and a centralized rule-maintenance mechanism.

16. Final Solution Statement

Solution 3 provides an implementation-ready design for automated IT ticket classification in ServiceNow. It defines the input data, flow stages, decision branches, classification mapping, validation approach, exception handling, and future extension points. The design directly supports the project objective of reducing manual ticket classification while maintaining a clear and manageable Flow Designer implementation.